

ISyE 6525 High Dimensional Data Analytics - HW1

Module 1: Introduction to High-Dimensional and Functional Data

Fall 2026

Submission. Submit *one PDF report per question*, and for every question that requires code, *one code file for that question*. You may use MATLAB, R or Python. Total: **100 points**.

Question 1.

(20 points)

The behavior of polynomials fit to data tends to be erratic near the boundaries. The polynomials fit beyond the boundary knots behave even more wildly than the corresponding global polynomials in that region. Assuming the function is linear near the boundaries (where we have less information anyway) is often considered reasonable. A *natural* quadratic spline adds this additional constraint, namely that the function is linear beyond the boundary knots. Let

$$y(x) = \begin{cases} \sum_{j=1}^3 \beta_j x^{j-1} + \sum_{k=1}^K \theta_k (x - \xi_k)_+^2, & x \in [\xi_1, \xi_K], \\ \sum_{j=1}^2 \beta_j x^{j-1} + \sum_{k=1}^K \theta_k (x - \xi_k)_+^2, & x \text{ outside } (\xi_1, \xi_K). \end{cases}$$

Determine a set of basis functions for $y(x)$ such that $\frac{d^2}{dx^2}y(x) = 0$ for x outside the interval (ξ_1, ξ_K) . Show your work.

Question 2.

(20 points)

Let $B_{i,j}(x)$ be the i^{th} B-spline basis function of a uniform quadratic B-spline with six knots. The B-spline curve is defined as

$$f(x) = \sum_{i=0}^2 B_{i,2}(x) P_i, \quad B_{i,0}(x) = \begin{cases} 1 & \tau_i \leq x < \tau_{i+1} \\ 0 & \text{otherwise} \end{cases}$$

$$B_{i,j}(x) = \frac{x - \tau_i}{\tau_{i+j} - \tau_i} B_{i,j-1}(x) + \frac{\tau_{i+j+1} - x}{\tau_{i+j+1} - \tau_{i+1}} B_{i+1,j-1}(x).$$

Derive an explicit piecewise expression for $B_{0,2}(x)$, $B_{1,2}(x)$ and $B_{2,2}(x)$. Note that *uniform* B-spline means the knots are equidistant, $\tau_{i+1} - \tau_i = \text{constant}$ for all i . For simplicity let $\tau_i = i$ (this is allowable because scaling or translating the knot vector has no effect on the shape of the $B_{i,j}$), so the knot vector is $X = \{\tau_0, \tau_1, \tau_2, \tau_3, \tau_4, \tau_5\} = \{0, 1, 2, 3, 4, 5\}$. Plot the three basis functions on the same axes.

Question 3.

(35 points)

Capacity fade in lithium-ion cells is monitored by repeatedly recording the terminal voltage during a constant-current discharge. **Question3.csv** contains the discharge profiles of 50 nominally identical

cells. Each row is one cell and each column is one of 200 equally spaced points of the normalized discharge time $t \in [0, 1]$; the entries are cell voltages in volts.

Compute the sample mean function $\bar{y}(t)$ of the 50 profiles and estimate it with the following four models. Report the cross-validated mean squared error (CV MSE) of all four on the *same* 5-fold split of the 200 grid points, so that the values are comparable. Select each tuning parameter as indicated below.

- (a) **(8 pts)** Cubic splines with the truncated power basis. Vary the number of interior knots from 5 to 30 (equally spaced), choose the best by 5-fold CV, and report the optimal number of knots and the CV MSE.
- (b) **(8 pts)** Cubic B-splines. Vary the number of interior knots from 5 to 30, choose the best by 5-fold CV, and report the optimal number of knots and the CV MSE.
- (c) **(8 pts)** Smoothing spline. Choose λ by generalized cross-validation, as on the slides. Report the optimal λ , the effective degrees of freedom $\text{tr}(\mathbf{S}_\lambda)$, and the CV MSE.
- (d) **(8 pts)** Kernel regression with a Gaussian kernel. Choose the bandwidth by 5-fold CV and report the optimal bandwidth and the CV MSE.
- (e) **(3 pts)** Collect the four CV MSE values in one table, plot the four fitted curves against $\bar{y}(t)$, and state which model you would select and why.

Question 4.

(25 points)

Tool wear in a milling operation changes the acoustic-emission (AE) signature of the cut. A worn insert releases a secondary burst of energy late in the cutting cycle that a sharp insert does not produce, but the burst is small compared to the cycle-to-cycle variability, so a single threshold on the raw signal does not work. The goal of this question is to classify a cut as *sharp* (class 0) or *worn* (class 1) by treating the AE traces as functional data.

`Question4_train.csv` contains 120 training signals and `Question4_test.csv` contains 60 test signals. In both files each row is one signal: the first 128 columns are the AE amplitude sampled on an equally spaced grid $t \in [0, 1]$, and the last column is the class label.

- Part 1. (10 pts)** Use cubic B-splines with 10 knots to reduce the dimension of each signal. Use the resulting B-spline coefficients as features to train a support vector machine with a Gaussian (RBF) kernel. Report the accuracy and the confusion matrix on the test set.
- Part 2. (12 pts)** Use functional PCA on the training signals for dimension reduction. Report the cumulative fraction of variance explained by the first ten harmonics and plot the first four estimated eigenfunctions. Build four SVM classifiers using 2, 5, 8 and 10 FPC-scores as features and report the accuracy and confusion matrix of each.
- Part 3. (3 pts)** Compare the two feature-extraction approaches. Explain in two or three sentences why the number of retained harmonics matters so much in Part 2.